

LD15M high intensity LED metallic beacon range

Ex db, weatherproof
Hazardous locations, ordinary locations



Overview

These certified beacons have been designed for use in potentially explosive atmospheres and harsh environmental conditions. The enclosures are suitable for use offshore or onshore.

The latest addition to the flagship beacon range aligns with the GRP universal base for greater application flexibility and simplified ordering.

Improved standard features include a customer-removable guard, upgraded mounting hardware, and swappable lens covers† for simplified maintenance.

The LD15 incorporates a low maintenance, innovative LED array, which has a life of up to 54,000 hours. The low current consumption design allows an energy efficient solution without compromising light output.

The housings are manufactured from 316L stainless steel or LM25TF marine-grade aluminium. Stainless steel screws and a mounting backstrap are incorporated ensuring durability in the field.

As well as the standard worldwide certificates, SIL 2 is available.

Certifications

- Zones 1, 2, 21 & 22
- Ex db IIC, T4/T5/T6
- ATEX certified, Ex II 2GD
- IECEx certified Gb, Db
- UL certified for USA and Canada:
 - Hazardous locations:
 - Class I, Div. 2, Groups A, B, C & D
 - Class II, Div 2 Groups F & G
 - Class I, Zone 1, AExd IIC T4/T5/T6
 - Ordinary locations: visual-signal device
 - Marine listed (UL ordlocs only)
 - Conforms to ULC regulated power supplies
- INMETRO certified
- SIL 2 certified
- IP66 & IP67

Features

- Certified temperature
 - ATEX/IECEX/INMETRO –55°C to +85°C (–67°F to +185°F)
 - UL –67°F to +158°F (–55°C to +70°C)*
- 5 user-selectable functions:
 - 60 fpm
 - 80 fpm
 - 120 fpm
 - Double flash
 - Steady
- Pipe mount or direct mount
- 316L stainless steel or LM25TF aluminium
- Optional remote initiate (telephone† or relay 24Vdc)
- Removable stainless steel backstrap
- Removable wire lens guard
- Consistent extreme temperature start-up
- Xenon version available (see datasheet dsmc0059 for XB15M)
- GRP version available (see datasheet dsmc0028 for LD15)

*For full temperature and T-ratings, see specification table

†Lens cover and telephone initiate is ATEX, IECEx and INMETRO only

Certifications	
ATEX Ex d	Cert. no. Baseefa 04ATEX0009X Certified to: EN60079-0, EN60079-1, EN60079-31 Ex II 2GD Ex db IIC T4/T5/T6 Gb, Ex tb IIIC T85°C/T100°C/T135°C Db
IECEX Ex d	Cert. no. IECEX BAS 05.0048X Certified to: IEC60079-0, IEC60079-1, IEC60079-31 Ex II 2GD Ex db IIC T4/T5/T6 Gb, Ex tb IIIC T85°C/T100°C/T135°C Db
INMETRO Ex db	Ex db IIC T4/T5/T6 Gb, Ex tb IIIC T135°C/T100°C/T85°C Db
UL	Cert. no. E187894 Class I, Div 2, Groups A, B, C & D Class II, Div 2, Groups F & G Class I, Zones 1, AExd IIC T4/T5/T6
SIL	SIL2 certified to IEC61508. Cert. number TBC.

Specifications	
Material	Body: stainless steel 316L or aluminium alloy LM25TF Lens: borosilicate glass Lens cover: polycarbonate (ATEX, IECEX and INMETRO only) Backstrap: stainless steel 316L Wire guard: stainless steel wire 316L Certification label: polymer
Finish	Cover ring may be painted to customer specification. Body is always black.
Lens colour	UL orders will be provided with a lacquered lens and excludes magenta. ATEX and IECEX orders will be provided with an interchangeable polycarbonate lens cover.
Voltage	12V d.c. to 48V d.c, 110V a.c. to 254V a.c
LED life	Up to 54,000 hours
Function	Steady, 60, 80 & 120 fpm, double flash
Certified temp ATEX, IECEX and INMETRO	-55°C to +85°C (T4) (-67°F to +185°F) -55°C to +55°C (T5) (-67°F to +131°F) -55°C to +40°C (T6) (-67°F to +104°F)
Certified temp UL	-55°C to +70°C (T5) (-67°F to +158°F) -55°C to +55°C (T6) (-67°F to +131°F)
Certified temp SIL 2	-50°C to +70°C (-58°F to +158°F)
Weight	Aluminium: 3kg; stainless steel: 5.25kg
Ingress protection	IP66 & IP67
Entries	Direct mount: 2x M20 with 1x SS316 plug (side) Direct mount: 2x ¾" NPT with 1x SS316 plug (side) Pipe mount: 2x ¾" NPT (side) + 1x ¾" NPT (base) with 2x SS316 plugs Pipe mount: 1x 3/4" NPT (bottom) Other entry configurations available upon request.
Terminals	Entries option A: 12 x 2.5mm²/14AWG Entries option C: 8 x 14AWG
Remote initiate	Telephone (ATEX,IECEX and INMETRO only): operates from ringing volt- age (default). Relay: operates with 24Vdc activation.
Certification labels	Certification labels always affixed to the base. IECEX, UL & CSA-approved high-durability, laser-marked polymer. Stainless steel 316 labels available as option 'C'.
Tag/duty labels	Optional. Tag & duty labels, if required, attach to the wire guard as swing tags and are packed separately to be attached on-site at point of installation. If tag and duty labels are required, please specify this when placing your order using the below part numbers: Single-sided tag: PX99904142 Single-sided duty: PX99904143 Double-sided duty/tag: PX99904144
EOL Resistor	Customers may fit a monitoring resistor to DC units. The maximum total power dissipation through the resistor must be limited to 2W. Please see the technical manual for further details.

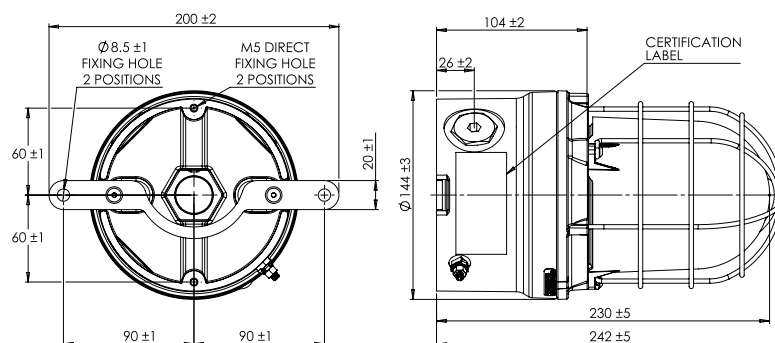
Electrical ratings					
Mode	Current at Nominal Voltage (mA)				
	12Vdc	24Vdc	48Vdc	120Vac (60Hz)	240Vac (50Hz)
Steady	430	220	120	80	40
60/80/120fpm	880	420	220	100	50
Double flash	710	340	180	100	50

Light outputs						
Typical light output - effective candela) (lacquered lens, no guard):						
Colour	Red	Blue	Green	Amber	White	Yellow
60 fpm	61	24	86	55	128	122
80 fpm	55	22	78	49	117	112
120 fpm	47	19	66	42	100	95
Steady†	41	17	59	39	86	82
DC: UL * 60/80/120fpm	25	5	14	18	48	32
AC: UL * 60/80/120fpm	12	6	13	8	68	37

*Tested to ULC-S526-07 requirements. Less than 15cd is not suitable for public mode.
†Not available UL

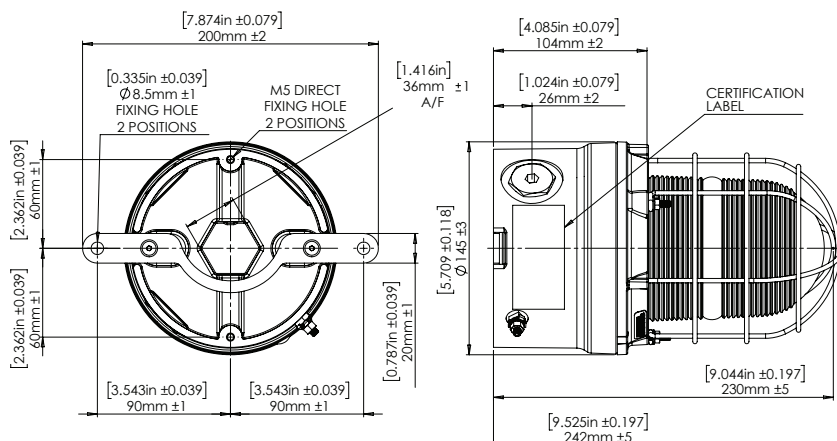
ATEX, IECEx and INMETRO certified universal base with lens cover

Certification codes: AJ, DM, W



UL certified universal base with lacquered lens

Certification codes: AJU, UL, UW



Ordering requirements

The following code is designed to help in selection of the correct unit. Build up the reference number by inserting the code for each component into the appropriate box.

Model	Material	Certification	Voltage	Lens colour	Lens guard	Entries	Options	Finish
LD15M					W			
Material Stainless steel S Aluminium A Certification Dual ATEX/IECEx AJ Triple ATEX/IECEx/UL AJU INMETRO DM UL UL UL(ordinary locations) UW Weatherproof W Voltage 12-48Vdc DC 110-254VAC AC Lens Red R Blue B Green G Amber A Yellow Y Clear C Guard Wire W Entries 2x M20 with 1x SS316 plug (side) A 2x 3/4"NPT (side) + 1x 3/4"NPT (base) with 2x SS316 plugs C 2x 3/4" NPT with 1x SS316 plug (side) E 1x 3/4" NPT (bottom) P Options None N SS316 cert labels C Relay initiate R*† Telephone initiate IT‡ EOL resistor ES Finish Black N Red R Special S*								
Note: finish applies to lens cover ring only, body is always black. *Please specify. * 24Vdc only † not available on SIL ‡ Available only for ATEX/IECEx/INMETRO. Refer to specifications table for details. § Please specify. If tag and duty labels are required, see specification table for full details.								
UL orders will be supplied with a lacquered lens and no lens cover.								
For any other configuration, please contact MEDC.								